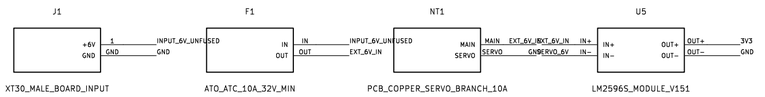
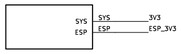
POWER / PROTECTION -- J1 board side is XT30 MALE; source cables FEMALE



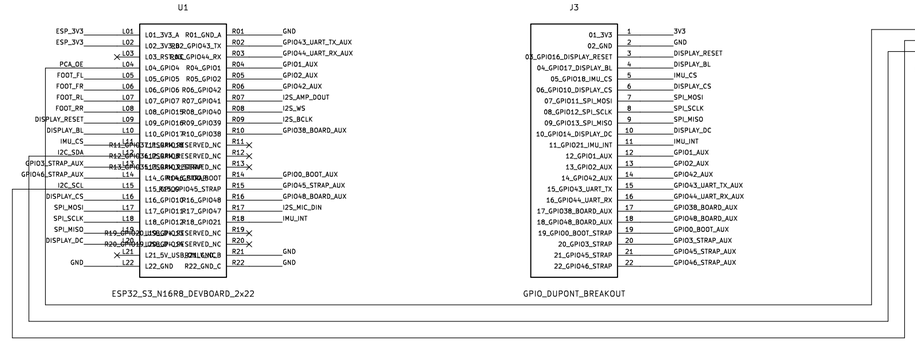
XT30_MALE_BOARD_INPUT | ATO_ATC_10A_32V_MIN | PCB_COPPER_SERVO_BRANCH_10A | LM2596S_MODULE_V151 | 1000-2200uF_16V_105C_OPTIONAL | 1000-2200uF_16V_105C_OPTIONAL | 10uF_16V_X7R | 100nF_16V_X7R

LM2596 INPUT CHECK: product spec VIN 3.2-40V and >=VOUT+1.5V; 6.0V-3.3V=2.7V PASS. Load-test at acceptance.

ESP32-S3 N16R8 candidate -- JP1 removable external-3V3 link; review USB/external power interaction



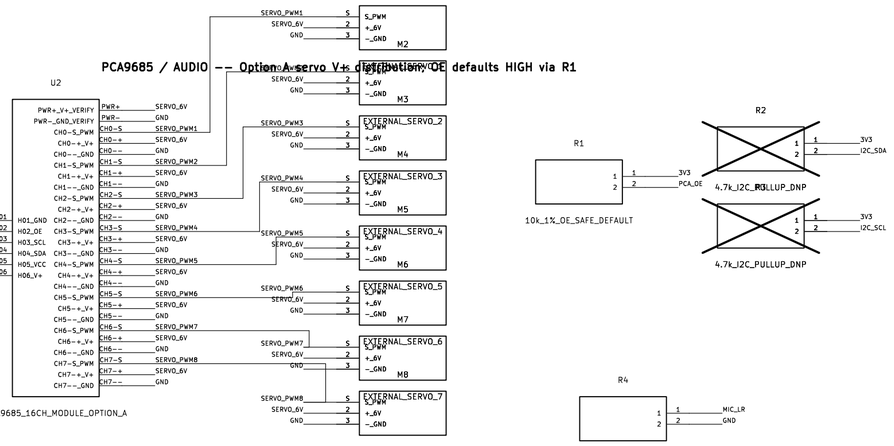
ESP_EXTERNAL_3V3_LINK_REMOVE_FOR_USB_ONLY_TEST



ESP32_S3_N16R8_DEVBOARD_2x22 | GPIO_DUPONT_BREAKOUT

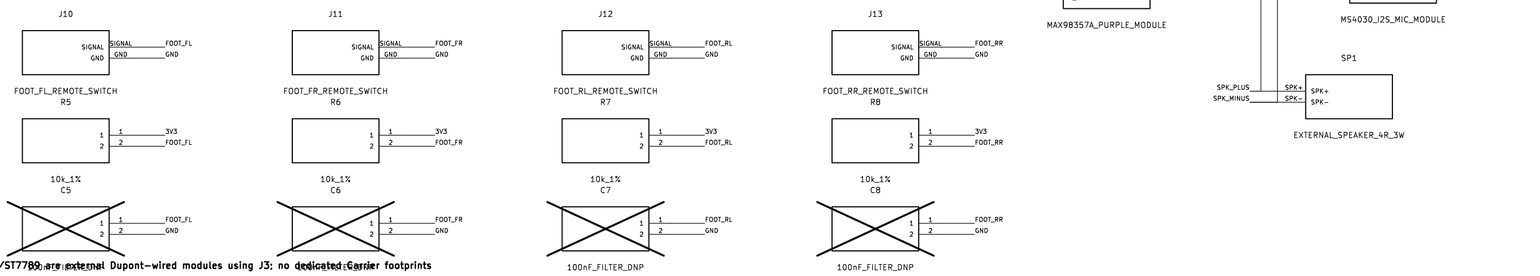
U1 is one complete 2x22 module: L01/R01 start at antenna end. J3 exposes fly-wire and unused GPIO.

PCA9685 / AUDIO -- Option A servo V+ distribution; OE defaults HIGH via R1

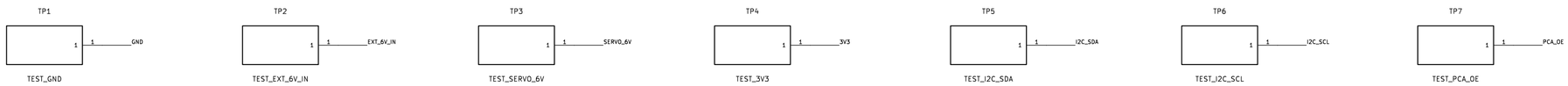


I2S clocks shared: GPIO39=BCLK, GPIO40=WS; GPIO41->amp DIN; GPIO47<-mic SD

FOOT INPUTS -- active LOW; 10k pull-up; optional 100nF DNP



MPU6500/ST7789 are external Dupont-wired modules using J3; no dedicated Carrier footprints



REVIEW CANDIDATE -- FOOTPRINT_TBD fields must be bound and imported nets checked in EasyEDA Pro

NOT READY FOR PCB: exact footprints and physical module pinouts pending
Single XT30 input / 10A fuse / ESP32-S3 / PCA9685

QuadQWOP

Sheet: /
File: quadqwop_carrier_v0.2.kicad_sch

Title: QuadQWOP Modular Carrier v0.2

Size: A2	Date: 2026-09-01	**Rev: Review Candidate 4 - p**
KiCad E.D.A. 10.0.3		Id: 1/1